

GenMol UDLM engineering exploration

engineering-v10-predictor-resolution | complete | 2026-09-07T11:40:48.728679+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

CT and clean CE share the training settings recorded in this protocol. Equal updates and examples do not imply equal compute or independently optimized objectives; training losses have different scales and targets.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED $\geq$ 0.6 and SA $\leq$ 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Prospective sampling design

```
{"compute_comparison": "512 predictor NFEs cost four times 128 per molecule; measure wall time and make no equal-compute or fixed-NFE claim.",
"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "nfe_by_configuration": {"ce_t050_n128": 128, "ce_t050_n512": 512, "ct_t050_n128": 128, "ct_t050_n512": 512}, "parameterization_controls": "Convert CE clean-denoiser logits to LOO at the current noisy state/time before temperature and top-p; CT logits already parameterize raw LOO.", "randomness": 0.0, "raw_loo_top_p": 1.0, "resolution_comparison": {"aggregation": "Equal-seed mean and sample SD; all declared seed pairs must define the metric before summary; undefined values are not zero.", "contrasts": [{"contrast_id": "ct_512_minus_128", "control_config": "ct_t050_n128", "method": "CT", "treatment_config": "ct_t050_n512"}, {"contrast_id": "ce_512_minus_128", "control_config": "ce_t050_n128", "method": "CE", "treatment_config": "ce_t050_n512"}], "decoding_branches": ["released_comparable", "strict"], "direction": "512 minus 128 within each method and seed; report both contrasts even when negative.", "metrics": ["validity", "uniqueness", "quality", "diversity"], "paired_unit": "Seed/run labels, not molecule-level trajectories."}, "selection_history": "V5/V6 and complete V9 outcomes were observed before this design. Temperature 0.5 is an informed engineering choice; both frozen CT and CE checkpoints remain included.",
"temperature": 0.5}
```

This design follows V5/V6 and complete V9 results; temperature 0.5 was selected using those outcomes and is not independent confirmation.

The 512-step treatment costs four times the predictor NFEs of 128, so any gain is not an equal-compute win; wall-clock scaling is measured.

The finite-state oracle uses exact untempered conditionals; it does not prove an improvement for learned temperature-0.5 molecular models or identify the cause of their quality gap.

The original V8 CT controller and campaign remain failed. This panel reuses its separately audited checkpoint plus the completed V8b CE checkpoint; neither is retrained or substituted.

Training source revisions differ through controller/reporting changes; model/training implementation and resolved common training settings match by recorded hashes.

128000 training exposures are configured accounting, not distinct molecules or an independent live row trace. Finite final tensors do not establish finiteness of every intermediate update.

Two seeds of 100 requests per setting are small; pairing is by seed/run, not molecule-level trajectories or independence. Withhold each contrast summary until both pairs define its metric.

Report validity, uniqueness, quality and diversity under strict SAFE decoding and released repair/largest-component selection, plus every per-seed 512-minus128 contrast. Quality counts unique QED $\geq$ 0.6 and SA $\leq$ 4 molecules divided by requests.

MDLM and paper means remain contextual comparisons with different sample/training budgets. Final UDLM seeds 0/1/2 remain reserved; no superiority, promotion or additional experiment is authorized by the results alone.

Objective comparison

CT / raw-LOO predicts clean leave-one-out probabilities. Clean CE predicts clean denoiser probabilities, converted to LOO logits at the current noisy state and time before temperature or top-p. The conversion adds no backbone evaluation. Shared sampling settings do not imply independently optimized objectives. Objective labels and checkpoint evidence below describe completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"CE": {"checkpoint_sha256": "b7d674ed5bddd1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020, "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b"}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}, "CT": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846, "original_v8_protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "prior_metadata_sha256": "f738b8b17de5c4704058018bddacd7fed779c85248e33d199b68a648151e612", "prior_variant":

"empirical_frequency", "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50},
"training_implementation_hashes_sha256": "c9641b7e447d1f4617599e4df338fde0010fe68f7877566de37d2e6c5a6ccd2b", "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdln_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
ct_t050_n128	1700	128	128	0
ct_t050_n128	1701	128	128	0
ct_t050_n512	1700	512	512	0
ct_t050_n512	1701	512	512	0
ce_t050_n128	1700	128	128	0
ce_t050_n128	1701	128	128	0
ce_t050_n512	1700	512	512	0
ce_t050_n512	1701	512	512	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
ct_t050_n128	1700	CT / raw-LOO	raw_loo	ema
ct_t050_n128	1701	CT / raw-LOO	raw_loo	ema
ct_t050_n512	1700	CT / raw-LOO	raw_loo	ema
ct_t050_n512	1701	CT / raw-LOO	raw_loo	ema

Config	Seed	Objective	Parameterization	Weights
ce_t050_n128	1700	clean CE	x0_denoiser	ema
ce_t050_n128	1701	clean CE	x0_denoiser	ema
ce_t050_n512	1700	clean CE	x0_denoiser	ema
ce_t050_n512	1701	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t050_n128	2/2	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.4550 +/- 0.0778	0.8918 +/- 0.0064
ct_t050_n512	2/2	0.9950 +/- 0.0071	1.0000 +/- 0.0000	0.4500 +/- 0.0283	0.8907 +/- 0.0011
ce_t050_n128	2/2	0.9900 +/- 0.0141	0.9949 +/- 0.0072	0.4650 +/- 0.0071	0.8928 +/- 0.0026
ce_t050_n512	2/2	0.9850 +/- 0.0071	1.0000 +/- 0.0000	0.4800 +/- 0.0424	0.8927 +/- 0.0011

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_t050_n128	2/2	0.7200 +/- 0.0990	1.0000 +/- 0.0000	0.3550 +/- 0.1202	0.8754 +/- 0.0016
ct_t050_n512	2/2	0.7350 +/- 0.0212	1.0000 +/- 0.0000	0.3250 +/- 0.0354	0.8764 +/- 0.0052
ce_t050_n128	2/2	0.7250 +/- 0.0636	1.0000 +/- 0.0000	0.3600 +/- 0.0000	0.8781 +/- 0.0007
ce_t050_n512	2/2	0.7150 +/- 0.0071	1.0000 +/- 0.0000	0.3850 +/- 0.0354	0.8779 +/- 0.0051

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
ct_t050_n128	1700	completed	100	100	0.4000	0.2700	15.14
ct_t050_n128	1701	completed	100	100	0.5100	0.4400	13.75

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
ct_t050_n512	1700	completed	100	100	0.4300	0.3000	57.41
ct_t050_n512	1701	completed	100	100	0.4700	0.3500	52.18
ce_t050_n128	1700	completed	100	100	0.4700	0.3600	14.95
ce_t050_n128	1701	completed	100	100	0.4600	0.3600	14.05
ce_t050_n512	1700	completed	100	100	0.5100	0.4100	63.67
ce_t050_n512	1701	completed	100	100	0.4500	0.3600	53.26

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={ 'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected { 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: cc6e3ced657f6ec66f611398e65b8b0e79598b3e76acbef77c72cfc4a7d0c4fd

v10-ct-t050-n128 / seed 1700: { "checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": { "diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": { "diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": { "ema": { "decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": { "cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": { "compute_mode": "Default", "compute_processes": [{ "pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid":

"GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n128/seed_1700/raw_samples.csv", "sha256": "c6abc1958624063f0bd300358d32d7a4d2ede63d90a989c95a17591577b18f5a", "size_bytes": 33423}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}
v10-ct-t050-n128 / seed 1701: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1309, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n128/seed_1701/raw_samples.csv", "sha256": "f4c29b56c1603787ff0d24435090db94b3b43f45b69d2da8bd97da381b19b20a", "size_bytes": 33293}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}
v10-ct-t050-n512 / seed 1700: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n512/seed_1700/raw_samples.csv", "sha256": "2cecd4957084f9510054f48a72ce65ec46520625a205b28c690424f3c7215e4a", "size_bytes": 34079}, "sampling_budget": {"corrector_steps_per_molecule": 0,

"gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}

v10-ct-t050-n512 / seed 1701: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1309, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 5}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ct-t050-n512/seed_1701/raw_samples.csv", "sha256": "e0482deaa9678cb658d05679c4d0688c5ceb7aa3dd1edcb4460cb9e49ace906", "size_bytes": 32825, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e"}

v10-ce-t050-n128 / seed 1700: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}, {"pid": 508363, "process_name": "/home/aidar.alimbayev/Documents/genmolv2/.venv/bin/python", "used_memory_mib": 2434}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 3749, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 5}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ce-t050-n128/seed_1700/raw_samples.csv", "sha256": "ab79badb11f5f808bc52c8ba6051d827e64ddc8bfd81d51ab71cca98d6baa938", "size_bytes": 34641, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v10-ce-t050-n128 / seed 1701: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ce-t050-n128/seed_1701/raw_samples.csv", "sha256": "aaafbe612da92ecefba46a04460ab18a4f53828e5ffd8f2f25a48037ce9aa0973", "size_bytes": 32214}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v10-ce-t050-n512 / seed 1700: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1286}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1309, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ce-t050-n512/seed_1700/raw_samples.csv", "sha256": "84ea6a2ffcc9644055669221cfa1c69a58cd5a81d68d1e74d5e3d7fca67d6c88", "size_bytes": 33955}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v10-ce-t050-n512 / seed 1701: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 395138, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1236}], "index": 7, "memory_total_mib": 49140, "memory_used_mib": 1259, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 7}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v10/v10-ce-t050-n512/seed_1701/raw_samples.csv", "sha256": "2035f6cb04278cf8d3c7950d3396be103a4d38cadeac21baa2ff44335a0cf7a8", "size_bytes": 32638}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "c17e848eb46896aff191ecd535d6a3cfa757c37e", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}